

Mamdouh Attia

Mobile Software Engineer | Flutter Specialist

[linkedin.com/in/mamdouh-attia/](https://www.linkedin.com/in/mamdouh-attia/) | github.com/Mamdouh-Attia | mamdouh-attia.github.io/mamdouh_portfolio/

✉ mamdouhattia31@gmail.com | +20-127-540-0234 | Egypt | Military Status: Completed

PROFESSIONAL SUMMARY

Flutter Developer with **+3 years** of hands-on experience building cross-platform mobile applications through **Clean Architecture** and **Agile** methodologies. Proficient in **state management** (Bloc/Cubit, Provider), Firebase, **REST API integration**, hardware sensors, and AI features. Open to remote/hybrid opportunities.

EDUCATION

Cairo University - Faculty of Engineering

Cairo, Egypt

Bachelor of Engineering - Computer Engineering Department

2019 – 2024

- Grade: Very Good with Honors
- Graduation Project Grade: Excellent

PROFESSIONAL EXPERIENCE

Flutter Developer (Open Source Contributor)

Feb. 2026 – Present

Itqan Community (mushaf-imad-flutter)

Remote

- Integrated **Quran.com API** via **OAuth 2.0**, reducing latency by **86%** via custom in-memory caching.
- Developed a reactive sync system with **Stream Transformers** for real-time verse metadata injection.
- Engineered a **Hybrid Audio Engine** supporting dynamic fallback between local and cloud sources.
- Authored **80+** tests with **Mocktail**, for asynchronous repository logic.

Flutter Developer

Sep. 2023 – Present

Freelance

Remote

- Developed stress monitoring app connecting to ESP8266 microcontroller via Wi-Fi with **real-time** sensor reading.
- Enhanced existing client app by adding Firebase authentication, push notifications, and QR code scanning.
- Integrated Blync API to enable real-time connection with hardware sensors in recycle bins.
- Utilized **MVC Architecture** with **Provider** state management.

Flutter Developer

Sep. 2023 – Jun. 2024

PrimaLab

New Damietta, Egypt | Hybrid

- Developed 2 client Flutter applications (VolunTeen and Check-It) handling all aspects of the SDLC.
- Built volunteering app targeting teenagers, featuring responsive 30–40 screens with AI chatbot integration.
- Developed mobile solution to verify product prices with barcode scanning and real-time price comparison.
- Integrated Firebase services, GPT-3 API, in-app video player, and third-party services.
- Delivered 50+ hours of hands-on Flutter training sessions covering Dart fundamentals, OOP, UI development, layout, widgets, and navigation with practical exercises and project-based learning.

Flutter Developer

Jun. 2023 – Sep. 2023

Orange Digital Center Software House

Cairo- Egypt | Hybrid

- Worked with agile teams on 2 internal mobile apps & 1 hackathon E-wallet app, meeting tight sprint deadlines.
- Translated Figma designs into Flutter UI and used MVVM architecture and Bloc Cubit.

PROJECTS

Mushaf-IMAD Package | Dart, REST API, OAuth2, Streams, Mocktail | PR #27 | Demo Feb. 2026 – Mar. 2026

- Integrated global cloud audio streaming with **Single-Trip** optimizations, reducing latency by **86%**.
- Designed a hybrid timing hierarchy (RAM → Local → API) ensuring **zero-latency** playback sync.
- Automated testing for complex async data models with **Mocktail**, achieving **80+** tests successful runs.
- Published architectural guides and walkthroughs to ensure long-term community maintainability.

PROJECTS

- EcoSort** | *Flutter, Freelance, Firebase, Blync API, QR, Push Notifications* Apr. 2024 – May 2024
- Enhanced an existing client app by adding Firebase authentication and push notifications.
 - Integrated Blync API to enable real-time connection with 3 hardware sensors in recycle bins.
 - Implemented QR code scanning and a point reward system for recycling activities.
- Check-It** | *Flutter, PrimaLab, Barcode, Provider, Forms* | [Portfolio](#) Feb. 2024 – Jun. 2024
- Developed a mobile solution to verify product prices and combat price manipulation in retail.
 - Implemented barcode scanning, real-time price comparison, and e-commerce style product database.
 - Enabled users to submit complaints and track their status.
- VolunTeen** | *Flutter, PrimaLab, Provider, Firebase, GPT-3* | [Portfolio](#) | [Demo Link](#) Sep. 2023 – Jan. 2024
- Solo project for a real client: built a volunteering app targeting teenagers, featuring responsive 30–40 screens.
 - Integrated AI chatbot (GPT-3), in-app video player, and third-party services.
 - Implemented MVC architecture, Firebase (storage), shared preferences, and WhatsApp/URL launcher.
- HyperMate** | *Flutter, Freelance, Wi-Fi service, ESP8266, GPT-3* | [Portfolio](#) | [Demo Link](#) Nov. 2023 – Dec. 2023
- Solo client project: developed a stress monitoring app connecting to ESP8266 microcontroller via Wi-Fi.
 - Displayed real-time GSR sensor readings and provided personalized exercise and nutrition suggestions.
 - Integrated a smart chatbot for user support and recommendations, and handled the Arduino firmware code.
- TabCash** | *Flutter, ODC's Hackathon, Agile, MVC* | [GitHub link](#) | [Presentation link](#) Apr. 2023 – May 2023
- Developed a fintech app prototype in a 2-week hackathon, showcasing idea of secure and easy transactions.
 - Built UI for home, wallet, statistics, and profile screens, and presented to Orange Digital Centre's board.
- Reddit Clone** | *Flutter, MVC-S, Agile, Provider, REST APIs* | [GitHub Link](#) | [Demo Link](#) Aug. 2022 – Dec. 2022
- Developed a Reddit mobile app clone as part of a 6-member team in a software house environment.
 - Implemented 3 main features : drawers, settings, and community creation screens.
 - Contributed to unit testing, mock services, documentation, code reviews and backend integration.
 - Followed Scrum Agile methodology with dedicated roles for frontend, backend, DevOps, and QA.

GRADUATION PROJECT

- Predicto** | *A+ | Python, OpenCV, Pytorch, Deep Learning, Desktop, Research* | [Demo video](#) Aug. 2023 – July 2024
- Developed an open-source Python library named 'vPredicto', specialized in VFP (Video Frames Prediction).
 - Researched and implemented from scratch the latest SOTA DL models in this field (SimVP - PredNet).
 - Responsible for GUI desktop app development using PyQt, integrating the 'vPredicto' library.

TECHNICAL SKILLS

- Programming:** Dart | C/C++ | Python | Java | SQL | HTML | CSS | JS | Go(basic)
- Mobile Development:** Flutter | Provider | Bloc/Cubit | MVC | MVVM | Clean Architecture | Streams | Shared Preferences | Flutter Web | Responsive & Adaptive Design | Animations
- Backend & Services:** Node.js | Firebase | Supabase | REST APIs | API Integration
- Hardware Support:** ESP8266 | Wi-Fi Service | Hardware Sensors | Arduino Firmware
- Features:** Push Notifications | QR Code & Barcode Scanning | AI Integration (GPT-3) | In-App Video Player | Searching & Filtering
- Testing & Quality:** TDD (Unit & Integration Testing) | Mock Services | Documentation | DevTools | Debugging
- Software Engineering:** SDLC | SOLID | OOP | Data Structures | Algorithms | Design Patterns | Clean Code | Dependency Injection | Asynchronous Programming
- Tools & DevOps:** Git/GitHub | GitHub Actions | GitHub Projects | Linux | Agile/SCRUM | Figma | AI-Assisted Development(GitHub Copilot - Stitch for UI design - Antigravity)

CERTIFICATES

- Career Essentials in GitHub Professional Certificate** | GitHub | [Certificate Link](#) 2026
- Machine Learning Specialization** | Coursera | [Certificate Link](#) 2024
- Flutter Internship Certificate** | ODC | [Certificate Link](#) 2023
- HCIA-AI** | Huawei | [Certificate](#) 2021